VeriSelf: Neutralizing AI Threats to Personal Privacy

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Abstract— Photos and videos can now be copied, transformed, republished and synthetically altered with very little effort. A protection system therefore needs to do more than run a single forgery classifier: it must keep a reference copy it can trust, recognise near-duplicates, flag possible manipulation, and present its findings in a form a human reviewer can understand. VeriSelf is a proposed privacy-conscious system for these tasks. Media is first registered with the owner's consent and stored securely, and cryptographic, perceptual and embedding representations are derived from it. Image analysis locates faces and passes them to an EfficientNet-B0 classifier; video analysis samples representative frames and models their temporal relationships. Authorized public-source monitoring uses similarity search to find candidate copies. All modules report to a central Core Platform that processes events, keeps them traceable, and sends notifications only after the evidence has been reviewed. Model outputs are treated strictly as evidence, never as an automatic verdict on ownership, alteration or misuse. This paper describes the architecture, processing workflow, security limits, evaluation plan and directions for future implementation.

Keywords— Media authentication, digital forensics, media fingerprinting, deepfake detection, RetinaFace, MTCNN, EfficientNet, similarity search, CNN-LSTM, privacy.

I. INTRODUCTION

Digital media is central to communication, education, entertainment, journalism and online social life. At the same time, the tools for editing and generating media have become more capable and easier to use. An image can be resized, recompressed, cropped or retouched, and a video can be edited frame by frame or have a face replaced. After such changes the original file and the copy found online may no longer share the same bytes.

Several distinct questions follow, and they are often confused. First, are two files exactly identical? Second, are they visually related even though their bytes differ? Third, does the related item contain manipulated content? Fourth, where does the candidate come from, and can it be linked to a previously registered object? No single score can answer all four, so complementary techniques are needed.

Strong building blocks already exist for the individual stages. FaceForensics++ offers a benchmark for manipulated faces [4]; RetinaFace and MTCNN provide face localization [1], [2]; EfficientNet gives an efficient convolutional backbone [3]; recurrent models add temporal reasoning for video [5], [6]; and FAISS supports large-scale vector similarity search [7]. Work on Face X-Ray, CNN-generated images, detector-evasion attacks and real-world generalization shows that detectors must be tested under varied conditions [8], [10], [11], [13], [14].

VeriSelf joins these pieces into one traceable workflow. When a media item is registered it receives a permanent identifier and several derived representations. Later candidates can be checked for exact identity, visual similarity or manipulation, and matches from permitted public sources can be retrieved by monitoring. The Core Platform links the evidence, records the processing context and routes relevant cases to human review. The aim is a media-authentication platform built from independent detectors, not a single detector.

Main contributions

- Secure, consent-aware registration of reference media before any monitoring takes place.
- Exact and approximate identification using SHA-256 hashes, perceptual fingerprints and embeddings.
- Separate image and video manipulation analysis, with spatial processing kept apart from temporal reasoning.
- Similarity-search monitoring restricted to authorized public sources.
- A Core Platform that gathers evidence, maintains traceability, and supports review and notification.
- Privacy and security controls: encryption, access control, data minimization, retention limits and human oversight.

II. RELATED WORK

A. Face localization

RetinaFace is a single-shot, multi-level method for locating faces and facial landmarks in unconstrained images [1], and MTCNN reaches a similar goal through cascaded convolutional stages that progressively detect and align faces [2]. Either can serve as the preprocessing stage in VeriSelf. A key design choice is to keep localization separate from the authenticity classifier so the two can be tested independently. This also clarifies failures: if no face is found, the classifier cannot be expected to give a useful result, and a successful detection says nothing about whether the content is authentic. Detection output is therefore logged as processing history, not as an authenticity judgement.

B. Benchmarks and generalization

FaceForensics++ is a widely used benchmark covering several facial manipulation methods with a standardized evaluation protocol [4]. Face X-Ray later studied blending boundaries as a cue for detecting a broader range of forgeries [8]. A practical

lesson is that a classifier should be one piece of evidence, particularly because deployment data can differ from training data. Detectors often behave differently when compression level, image source or post-processing changes [10], [11], and detector-evasion attacks show that strong results in one setup do not imply resistance to deliberate attacks [13]. VeriSelf therefore plans robustness testing from the start and does not link the classifier to identity, source or ownership; it reports its prediction with the region processed and the model configuration.

C. Efficient convolutional networks

EfficientNet balances network depth, width and input resolution through compound scaling [3]. VeriSelf adopts EfficientNet-B0 as a lightweight backbone for image-level analysis. This is an architectural choice, not a claim that B0 is optimal; larger variants or other backbones can be compared later behind the same service interface.

D. Video manipulation detection

Video adds a temporal dimension that a single image lacks. Sabir et al. studied recurrent convolutional approaches for detecting manipulated faces in video [5], and the DeepFake Detection Challenge dataset shows the scale and diversity such detectors must handle [9]. These findings motivate a pipeline that samples representative frames, extracts spatial features and then models the sequence. The temporal stage complements frame-level analysis: an artifact that is hard to see in one frame may be clearer across consecutive frames of the same face region.

E. Similarity search and provenance

FAISS performs efficient nearest-neighbor search over large sets of high-dimensional vectors [7], which matters because directly comparing every candidate with every registered item would be too expensive. A vector index lowers retrieval cost and leaves capacity for deeper analysis and review. On the provenance side, the C2PA specification defines cryptographically verifiable, tamper-evident content credentials [12]. VeriSelf can store such credentials as an additional evidence channel; provenance metadata and similarity search answer different questions and should not be used interchangeably.

III. PROPOSED VERISELF FRAMEWORK

The framework has six functional areas: media registration, media fingerprinting, image manipulation detection, video manipulation detection, authorized monitoring, and notification and response. A central Core Platform coordinates them and maintains the links between a registered item, its derived representations and every analysis performed on it.

Fig. 1. Overall VeriSelf architecture and information flow.

The Core Platform is not itself a detection model. It receives a registration event, calls the analysis services it needs, and ties their results to a common media identifier. This keeps individual services from needing full access to user data, and it provides one place for timestamps, model versions, preprocessing settings and review status. The combined record supports review, but it should not be read as a mathematically proven ownership decision: a reviewer must still consider context, permissions and the source of the candidate media.

IV. SECURE MEDIA REGISTRATION

Registration creates the reference object on which everything else depends. The user uploads an image or video through an authenticated interface, and the consent and monitoring preferences are recorded before any derived representation is generated. The stored object and its metadata must be trustworthy, because later comparisons are only meaningful if the reference is.

Fig. 2. Secure media registration process.

The registration service collects the required consent and monitoring information, accepts only supported formats, checks integrity, and stores the original with access limited by deployment policy. It then follows two rules:

- Create perceptual and cryptographic representations only after the reference object has been accepted.
- Link the representations, metadata and retention state to a single registration identifier.

Later analysis starts from this record. SHA-256 provides exact binary identity, while perceptual and embedding representations are designed to survive selected visual transformations. For video, representative frames can be indexed rather than every frame, which avoids unnecessary comparison cost.

V. MEDIA FINGERPRINTING

No single fingerprint suits every verification task, so VeriSelf uses three complementary representations: a cryptographic hash to test whether two files are byte-identical, a perceptual fingerprint that captures visual properties and tolerates resizing or compression, and an embedding that allows flexible approximate retrieval from a large collection.

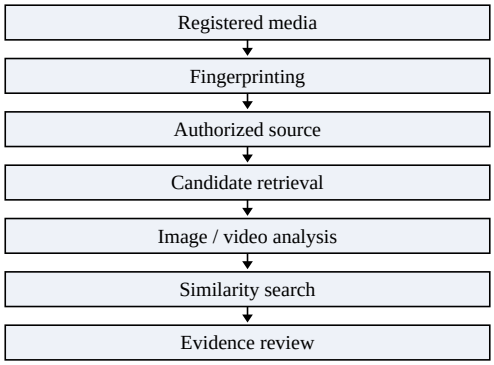

Fig. 3. Workflow for fingerprinting, retrieval and evidence handling.

A. Cryptographic reference

A SHA-256 digest is computed for each registered file. Because it depends on the full byte sequence, even a one-bit change produces a different digest. The hash is thus a tool for exact identity only; benign changes such as recompression or resizing will alter it, and it says nothing about visual similarity.

B. Perceptual representation

A perceptual hash captures selected visual characteristics rather than the exact bytes, so related images stay similar after common transformations. The matching threshold should be set experimentally for the chosen representation and dataset, not assumed to hold in every situation.

C. Embedding-based retrieval

An embedding maps media to a numerical feature vector that can be indexed for approximate nearest-neighbor search. Retrieval returns a short list of candidates for more expensive analysis, so manipulation detection need not run on every object in a large collection. A retrieval score is only a candidate indicator and is stored with the evidence record.

D. Evidence handling

Every comparison should record the representation type, model version, similarity metric, threshold and processing time against the registration identifier. This context is needed when a result is reviewed later or when a model changes. A candidate remains a reviewable event until its evidence has been interpreted in context.

VI. IMAGE MANIPULATION DETECTION

The image pipeline concentrates on face regions. RetinaFace or MTCNN locates faces and landmarks, the selected regions are normalized, and they are passed to an EfficientNet-B0 classifier. The service returns a class label together with the processing configuration needed to reproduce the result.

Fig. 4. Proposed pipeline for detecting image manipulation.

An event record should hold the predicted class and confidence, the analyzed region, the model version and a timestamp, which is easier to inspect than a bare binary label. A confidence value is a model estimate and does not by itself show that manipulation occurred. Cropping to the face reduces background variation and focuses the classifier where manipulation is expected; the crop and normalization policy should stay fixed during evaluation so preprocessing changes do not hide as performance differences. When a result is unclear, the system can request further analysis or route the event to a reviewer instead of forcing a decision.

VII. VIDEO MANIPULATION DETECTION

The video pipeline extends the image flow with changes in visual features over time. Rather than process every frame, the system samples representative frames, detects faces, extracts spatial features with a CNN and passes the resulting sequence to an LSTM or another temporal model.

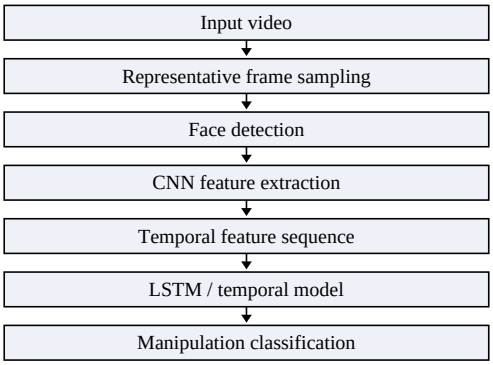

Fig. 5. The proposed video manipulation detection pipeline.

Sampling density is a key trade-off: dense sampling raises inference cost, while sparse sampling lowers cost but can miss short-lived artifacts. VeriSelf records the sampling interval, sequence type, frame resolution and temporal model with each result so evaluations remain reproducible. If several consecutive frames show consistent evidence, the event may deserve closer inspection, and keeping those frames as supporting material is more useful to a reviewer than reducing the whole video to one opaque score.

VIII. AUTHORIZED WEB MONITORING

The monitoring module only uses sources that a deployment is permitted to access. Registered media is converted into a

searchable representation, candidates are collected from authorized public sources, and candidate embeddings are compared with the registered ones.

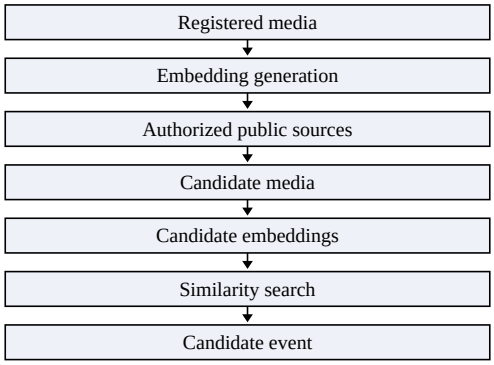

Fig. 6. Authorized workflow for monitoring purposes from public sources.

Retrieval can use FAISS or another vector-search system [7] and should be treated as a filtering stage. Similar appearance can come from legitimate reuse of material or shared backgrounds, so the monitoring service passes the source information, similarity scores and candidate references to review instead of automatically declaring misuse.

IX. NOTIFICATION AND RESPONSE

Sending a notification turns automated analysis into an operational event, so VeriSelf separates evidence generation from the action taken as a result. A typical path runs from analysis result, through confidence and evidence assessment, to a potential match or manipulation event, human review, notification and response. Notifications should state what the system detected without presenting a prediction as certainty, and should include the registered media identifier, candidate source, retrieval score, manipulation indicators, model version and processing time, so reviewers can read the alert without access to the original services. Events can be prioritized by similarity, strength of manipulation evidence, source information and whether the candidate matches a protected registration; this aids workflow and must not replace the ownership or misuse decision. Review outcomes are recorded as confirmed, rejected, inconclusive or still under analysis.

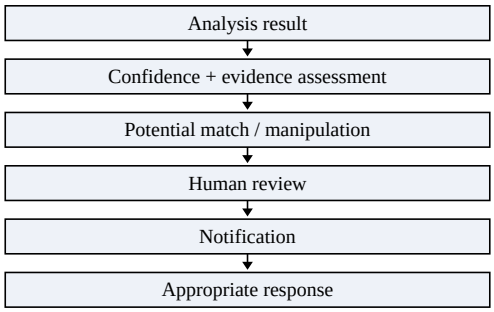

Fig. 7. Workflow for notification and response.

X. SYSTEM WORKFLOW AND ALGORITHMS

The end-to-end process connects specialized services without exposing all data to each one. Registration establishes the reference identity, fingerprinting supplies exact and approximate representations, the image and video modules assess manipulation, and monitoring identifies candidates. The Core Platform combines these outputs into one reviewable event. Because the services are modular, a detector can be upgraded without changing registration, and each part can scale independently, provided every result stays tied to the same registration ID and carries enough version data to reproduce the event.

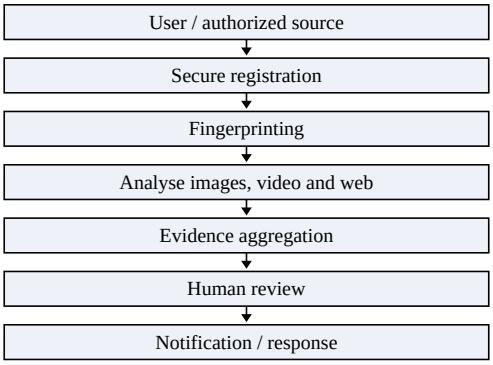

Fig. 8. End-to-end VeriSelf workflow.

Algorithm 1: Secure Media Registration

- Authenticate the user and record consent and policy settings.
- Check media type and integrity; encrypt and store the reference.
- Create the SHA-256 and perceptual/video representations.
- Keep only the metadata needed for permitted services and apply the configured retention policy.
- Return the registration identifier and protected-reference information.

Algorithm 2: Candidate Monitoring and Analysis

- Create embeddings for registered media and for candidates from authorized sources.
- Compute similarity and keep candidates that meet the retrieval policy.
- Run manipulation analysis if more evidence is needed.
- Package source, similarity and model evidence into an event for human review.
- Record the review result and any notification.

XI. SECURITY AND PRIVACY

Privacy is part of the architecture, not an add-on. Registered media may hold sensitive personal information, and derived embeddings, thumbnails, logs and reviewer notes can also reveal it, so controls cover the whole processing chain:

- Encryption: protect registered media in storage and in transit.
- Consent: process media only as permitted by its registration.
- Data minimization: give each service only what its task requires.
- Retention control: delete media and metadata when the retention period ends.
- Access control: limit protected data to authorized users and services.
- Authorized monitoring: use only sources the deployment may access.
- Human review: use automated output to support decisions, not to make consequential ones silently.

A deployment must also protect the services that process media. An attacker might replace a registered file, submit misleading metadata, alter derived representations, evade a

detector, or interfere with monitoring and notification. Fingerprints and embeddings are therefore bound to the registration identifier, so a substituted representation cannot quietly become the trusted reference, and access to the registration store is authenticated and logged. A manipulator may also use compression, resizing, cropping, filtering or frame removal to reduce detector reliability [10], [13]; VeriSelf answers this with controlled robustness testing and by treating predictions as evidence rather than truth. Model and preprocessing versions travel with every result so historical evidence stays interpretable after upgrades.

XII. EVALUATION PLAN

This paper presents a framework and has not yet measured a completed implementation, so it reports no performance figures. Evaluation is planned at two levels: each module in a controlled setting, and end to end from registration to notification.

A. Image and video detection

Image experiments can use a benchmark such as FaceForensics++ [4] and report accuracy, precision, recall, F1-score and ROC-AUC, along with the dataset split and preprocessing configuration. Testing should cover more than one manipulation type and add compression and resizing to study robustness against a clean test set. Video experiments should record the sampling interval, sequence length, temporal model, frame resolution and split, and may report frame-level or clip-level metrics under one fixed protocol so differences are not caused by inconsistent preprocessing.

B. Fingerprinting and similarity search

Table I. Fingerprinting test cases and expected behavior.

Test case	Expected behavior
Original file	Exact match
Resized image	Perceptual similarity
Compressed image	Perceptual similarity
Cropped image	Partial similarity identified
Unrelated image	Low similarity

The fingerprinting experiment should show how each representation behaves under transformations that change the bytes without removing visual content, and should be used to choose thresholds on validation data. Monitoring can be assessed with Precision@K, Recall@K, search delay, candidate counts and false-positive rate, and end-to-end tests should time registration, fingerprinting, retrieval, manipulation analysis and alert generation to find computational bottlenecks.

Table II. Suggested reporting format for the completed implementation.

Component	Metric	Result
Face detection	Precision	To be measured
Image manipulation	Accuracy / F1	To be measured
Video manipulation	Accuracy	To be measured
Fingerprinting	Match rate	To be measured
Similarity search	Recall@K / latency	To be measured

Reports should separate measured observations from architectural expectations. Models compared with one another should use the same dataset split, preprocessing policy and metrics, and results should be given for several media conditions so that a high score on one clean set is not mistaken for general reliability.

XIII. DISCUSSION, LIMITATIONS AND FUTURE WORK

The central design decision in VeriSelf is to use complementary evidence. A cryptographic hash gives exact identity, perceptual fingerprints tolerate some visual change, embeddings support large-scale retrieval, and manipulation models add a further kind of evidence. Together they let the system ask different questions at different levels. A manipulation score is not an ownership score and is not proof of misuse; by keeping source information, model outputs and processing history together, the reviewer can see why an alert was raised, which matters most where a wrong automatic response would have serious consequences. Detector or preprocessing updates should always be linked to the version that produced each historical result, or old evidence becomes hard to interpret.

- Evaluate the full workflow on larger and more varied image and video collections.
- Test robustness to compression, resizing and cropping, and to manipulation methods and transcoding not seen in training.
- Compare alternative temporal models for video and report the trade-off between accuracy and computation.
- Add confidence calibration and uncertainty estimation so model outputs better support review decisions without hiding uncertainty.
- Include provenance and content-credential data when reliable metadata is available.
- Measure run time and scalability as the number of registered items grows.

Because the Core Platform separates orchestration from the analysis services, new detectors, retrieval methods or provenance sources can be assessed without redesigning the registration workflow.

XIV. CONCLUSION

VeriSelf offers an integrated approach to media authentication and manipulation analysis, in which registration, fingerprinting, detection and monitoring are linked to response through a shared evidence layer. Exact-file verification is combined with perceptual and embedding-based comparison, and image or video manipulation analysis is added when a candidate needs deeper inspection. Automated analysis is kept separate from final human interpretation, model scores, similarity values and provenance data are preserved as evidence, and consent, access control, data minimization, retention policies and authorized monitoring are treated as architectural requirements.

This work describes a framework and an evaluation methodology, not measured performance. The next step is an implementation with reproducible preprocessing, clearly defined datasets, robustness tests and quantitative reporting, which will show where the workflow is useful in practice and where the models or system need improvement.

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